Topic 2 — Indus Valley Civilization (Harappan Civilization)

Covers syllabus (click to expand)

Indus Valley Civilization | Features of Harappan Civilization | Major Harappan Sites | Archaeological Sites and Excavations | Archaeological Sites of Uttar Pradesh | Major Ports of Ancient India | Harappan Economy | Harappan Trade | Town Planning | Drainage System | Agriculture | Craft Industries | Religion | Script | Seals | Weights and Measures | Dholavira | Rakhigarhi | Lothal Dockyard | Kalibangan | Banawali | Surkotada | Daimabad | Copper Hoard Culture | Ochre Coloured Pottery (OCP)

Sources baked in: NCERT *Themes in Indian History* I, RS Sharma, ASI, Ghatnachakra Indus Valley (B–17+), UPPCS Prelims PYQs 2018–2025, UKPCS / sister papers **Weight:** ★★★ — site↔state, UP trio, eastern boundary, dockyard/boats, excavators **Last verified:** September 2026 **Current Affairs:** Dholavira UNESCO **2021**; Rakhigarhi 2026 ASI DNA/facial reconstruction; Budget 2026 lists Lothal and Rakhigarhi among 15 experiential sites

Current Affairs

- **2021:** Dholavira → UNESCO World Heritage Site.
- **2018:** Sanauli chariot burials (ASI).
- **2026:** Rakhigarhi skeletons to AnSI (Kolkata) and BSIP Lucknow for DNA / facial reconstruction; ASI 3-year excavation.
- **Budget 2026:** 15 archaeological sites as experiential destinations, including **Lothal** and **Rakhigarhi** (राखीगढ़ी).

2.1 Indus Valley Civilization

Bronze Age urban culture | Mature ~2600–1900 BCE | also called Harappan | proto-historic

- First urban civilization of the subcontinent; contemporary with Mesopotamia and Egypt.
- Named **Harappan** after **Harappa** (हड़प्पा) (first excavated site). Geography is wider than the Indus: Ghaggar-Hakra (Saraswati belt), Gujarat, Haryana, eastern UP, northern Afghanistan.
- Archaeologists use **culture** for a set of distinctive objects found together in place and time. For Harappa these include **seals, beads, weights, stone blades, and baked bricks**. The **seal** is the most distinctive artefact.
- IVC is **proto-historic**: people wrote, but the script is still **undeciphered**. It is **not** prehistoric (no writing) and **not** fully historic (readable texts).
- Some books say **Sindhu–Saraswati** civilization — same culture, different label.

- **Early Harappan** (~3300–2600 BCE): regional cultures (Kot Diji, Amri, Hakra, Sothi). Pre-urban, pottery, mud-brick.
- **Mature Harappan** (~2600–1900 BCE): planned cities, seals, weights, long-distance trade. This is what papers mean by "IVC."
- **Late Harappan** (~1900–1300 BCE) is the de-urbanisation phase.
- Cemetery-H, Jhukar, and Lustrous Red Ware at **Rangpur** mark Late Harappan regional faces.
- **OCP** overlap appears in the doab in this late band.
- Booklet C-14 framing often quotes about **2300–1700 BCE**, with peak about **2300–1900 BCE**. Standard teaching still uses Mature **~2600–1900 BCE**. Older writer brackets (Marshall, Mackay, Vats, Gadd, Wheeler, Fairservis) differ; do not treat any one as absolute.
- Spread about **12.5–13 lakh km²** — larger than ancient Egypt or Mesopotamia in area.
- **West:** Sutkagen-dor (Makran / Dasht coast).
- **East:** Alamgirpur (Meerut, UP) on the **Hindon**.
- **North:** Manda (Jammu, on the Chenab).
- **South:** Daimabad (Maharashtra) on the **Pravara**.
- About **1400+** sites are known. Under **100** are fully excavated.
- The core cluster sits in Pakistan plus Haryana–Rajasthan–Gujarat.
- Origin is **indigenous** growth from Mehrgarh / Early Harappan. **Amalananda Ghosh** reads pre-Harappan maturing into Harappan. **Mackay** / some older views stressed Sumer migration; **Wheeler** stressed idea-migration from West Asia. **M. Rafique Mughal** argued local growth in the Ravi–Harappa region and **rejected** a Mesopotamian-inspiration model — so "Mughal = Mesopotamia-inspired" is the **wrong** match in origin stems.
- Older racial typology lists Proto-Australoid, Mediterranean, Alpine, and Mongoloid elements at Mohenjo-daro. Treat that as a **booklet race-list**, not modern genetics.
- Founder theories in older books: **Laxman Swaroop** (Aryan), **Childe / Wheeler** (Sumerian link), **R.D. Banerji** (Dravidian). None is a closed scientific proof.
- **Stuart Piggott** called Harappa and Mohenjo-daro the **"twin capitals"** of an extensive kingdom — a useful phrase, not a proven dual capital constitution.
- Decline starts about **1900 BCE**. Causes include climate shift, Ghaggar drying, floods, and overuse of land.
- **Flood** decline readings are linked with Marshall, Mackay, and S.R. Rao.
- **Aryan attack** readings by Childe, Wheeler, and Piggott are now **rejected** as a massacre model.
- **Climate** decline readings are linked with Aurel Stein and Amalananda Ghosh.
- **Geological change** readings are linked with M.R. Sahni, Lambrick, and G.F. Dales.
- **Epidemic** readings are linked with K.U.R. Kennedy.
- An IIT Kharagpur drought study is often cited for about **900** years of weak monsoon.
- Economy type in match-lists is **Urban** (not pastoral). Pastoral = Rigvedic.
- Compared with Vedic culture: IVC is **urban**, has a **pictographic** unread script, and lacks **iron**; Vedic society is rural / pastoral to agrarian and later uses iron.

> **Logic:** Eastern boundary = **Alamgirpur**, not Rakhigarhi. IVC economy type = **Urban** (Rigvedic = pastoral). Marshall's public announcement year is **1924** (excavations begin 1921–22).

> [!TIP] **Exam Anchor [UPPCS Prelims 2020, Q21]:** IVC = Urban; Later Vedic = Agrarian; Rigvedic = Pastoral; Medieval = Land Lordism. *(See Q5 in Complete PYQ Bank below)*

2.2 Features of Harappan Civilization

- Harappan cities are **planned** on a grid, usually with a western **citadel** and an eastern **lower town**. Dholavira adds a distinct **middle town**.
- Builders use **burnt bricks** in the ratio **1: 2: 4** (thickness: width: length) across a wide geography from Punjab to Gujarat. Ornamental bricks at Kalibangan and curved bricks at Chanhudaro show that “exactly one brick size everywhere” is false.
- Most houses are single-storey, and **two-storey** houses with stairs also exist.
- Main roads are often about **10 m** wide (Mohenjo-daro’s main street is about **9.15 m**).
- Cities use **covered drains**, house bathrooms, and wells. Drainage is best at Mohenjo-daro and **weak at Kalibangan**.
- Metal work is **bronze** (copper plus tin), including lost-wax casting. There is **no iron** and **no coined money**.
- Steatite **seals** and wheel-made **red** pottery (plain or black-on-red painted) are common. Polychrome pottery is rare.
- The script is still **undeciphered** (about 400–600 signs).
- Famous public works include the **Great Bath** (only at Mohenjo-daro), granaries at Harappa and Mohenjo-daro, and the dockyard at Lothal.
- There is **no** securely identified palace or stone temple, and no royal pyramid-tombs.
- Relatively few weapons appear in the record, though some sites are still **fortified** (Dholavira, Surkotada, Kalibangan citadel).
- Cotton is among the **earliest** in the world (often cited from about **3000 BCE** in India; Greek *Sindon* recalls Indus cotton). Standard stone weights use a binary system with key unit **16**.

> **Logic:** Iron plough / Ashokan-style temple / gold coins are **not** Harappan.

2.26 Social Organisation and Polity

- Harappan society was **urban and specialised**, but archaeologists have **not** identified a palace of a named king.
- The **"Priest-King"** steatite bust is an **interpretation**, not a proven title.
- **Merchants, seal-cutters, bead-makers, and bronze-smiths** formed a craft hierarchy fed by **agricultural surplus** in granaries.
- There is **no coined money**; exchange used **barter** and the **weight system** (key unit 16).
- Fortified sites show **defence**, but few weapons — the "peaceful civilisation" claim is partly true, partly oversimplified.

2.27 Early Harappan and Late Harappan Phases

- **Early Harappan** (about 3300–2600 BCE) is the pre-urban horizon of sites such as **Kot Diji**, **Amri**, **Hakra**, and **Sothi**.
- **Mature Harappan** (about 2600–1900 BCE) is the full urban phase of planned cities, seals, script, the Great Bath, and the Lothal dock.
- **Late Harappan** (about 1900–1300 BCE) is the age of **de-urbanisation**, with **Cemetery H** at Harappa and **Jhukar** and **Rangpur** ware horizons.
- In the eastern doab, Late Harappan overlaps **OCP** and then **PGW** — that sequence is not Mature Harappan urbanism.

2.3 Major Harappan Sites

Site ↔ river/state ↔ one signature find | Pakistan first, then India

- **Harappa** stands on the left bank of the **Ravi** (रावी) in Pakistani Punjab (Montgomery / Sahiwal district).
- **Daya Ram Sahni** excavated it in **1921**.
- **Stuart Piggott** paired it with Mohenjo-daro as **“twin capitals.”**
- It has two main mounds: a western **citadel** mound and an eastern **city** mound.
- Granary remains are often described as **twelve** chambers in **two rows of six**.
- Cemetery **R-37** includes coffin burial.
- The first famous artefact type is a **unicorn** seal with inscription.
- Finds include a red-sandstone **male torso**, bronze mirror, **6 granaries in a row** with working floors & workmen's quarters, virgin-goddess seal, cemetery **H** alongside R-37, stone symbols of lingam and yoni, wheat and barley in wooden mortar, copper vanity box, and dice.
- **Mohenjo-daro** means “mound of the dead” (also called **Nakhlistan** or “Oasis of Sindh”), largest site of the civilization, and lies on the right bank of the **Indus** (सिंधु) in **Larkana, Sindh**.
- **R.D. Banerji** excavated it in **1922**.
- The **Great Bath** is here. No other Harappan city has this exact public tank.
- A large building north-east of the bath (about **70 × 24 m**) is often read as a priest’s residence or college of priests.
- The main street (sometimes called **Rajpath**) is about **9.15 m** wide; streets meet at right angles.
- The bronze **Dancing Girl** is about **10.5 cm** and was cast by the lost-wax method.
- The **Priest-king** is a steatite bust with a trefoil shawl.
- The famous **humped bull** seal and **Pashupati** seal are from this city.
- Boat models / boat seals occur here. Cotton cloth evidence occurs here. Population is often cited at about **35,000–40,000**.
- Yielded **1,398 seals** (~56% of total Indus seals found), the Great Granary (largest building), assembly hall, and steatite bearded priest.
- **Chanhudaro** in Sindh is a **bead factory** town with **no citadel**, about **130 km** south of Mohenjo-daro.

- **N.G. Majumdar** discovered it in **1931**; **E.J.H. Mackay** excavated it on a large scale in **1935–36**.
- Crafts include shell work, bangles, and beads. Booklet extras include an inkpot, dog-following-cat track (dog's paw imprint on brick), bronze toy cart, and cosmetic / “lipstick” finds.
- **Jhukar** and **Jhangar** late cultures also appear here.
- **Sutkagen-dor** on the Makran is the **western** sea-gate of the culture.
- **Sotka Koh** is the neighbouring Makran coastal site.
- **Kot Diji** in Sindh is an Early Harappan / pre-urban site.
- **Amri** is another Early Harappan site in Sindh.
- **Nausharo** is an Early-to-Mature sequence site.
- **Rehman Dheri** is a pre-urban planned settlement in the Gomal belt.
- **Gumla** is an Early Harappan site in the same northwest cluster.
- **Balakot** is a coastal site known for shell and fish.
- **Alladino** is another coastal craft site.

India

- **Dholavira** stands on Khadir island in **Kutch**.
- It is known for giant water **reservoirs**, unique stormwater drainage, a **10-sign board** (civic inscription), and a ceremonial ground / **stadium**.
- The city has **three** divisions and more **stone** architecture than the Indus brick cities.
- UNESCO inscribed it in **2021**.
- **Lothal** lies in the Ahmedabad belt on the **Bhogavo**.
- **S.R. Rao** excavated it.
- The **dockyard** is about **218 × 37 m**.
- A warehouse stood beside the dock.
- Lothal had a bead factory and metal / shell workshops.
- Rice husk is reported here.
- A fire altar occurs at Lothal.
- Boat models / ship terracotta also occur at Lothal.
- An **ivory scale**, Persian Gulf and Bahrain seals, terracotta horse figurine, dyeing vat, and painted jar depicting bird and fox / thirsty crow story are booklet extras for Lothal.
- **Double burial** evidence is reported here (also at Kalibangan and Rakhigarhi in some reports).
- **Surkotada** in Kutch is a fortified settlement.
- **J.P. Joshi** discovered and excavated it.
- **Horse bones** and pot burials are reported. The horse identification is **disputed**.
- **Rangpur** in Saurashtra (Gujarat) is a Late Harappan site with lustrous red ware.
- Vegetal remains support **rice, bajra, and sorghum** cultivation claims.
- **Rojdi** in Gujarat is a Sorath Harappan settlement.
- **Desalpur** in Kutch is a fortified Harappan site.

- **Kuntasi** in Gujarat is a coastal craft and port-linked site.
- **Padri** in Gujarat was the 2020 match for **Gujarat**.
- **Bhagatrav** in Gujarat is a coastal Harappan site.
- **Kalibangan** in Hanumangarh, Rajasthan, stands on the **Ghaggar** (घग्गर).
- **B.B. Lal** and **B.K. Thapar** excavated it (from about **1960–61**). Luigi Pio Tessitori and later **A. Ghosh** also figure in early notice / survey.
- Name means “black bangles”. It is known for a pre-Harappan **ploughed field**, **seven fire altars**, and decorated bricks.
- A triangular terracotta cake with a **horned deity** is a Kalibangan booklet fact.
- Terracotta cakes and camel bones occur. Drains are poorer than at Mohenjo-daro.
- Mesopotamian-type seals and toy-cart wheels are also reported in booklet notes.
- **Banawali** in Fatehabad, Haryana, was excavated by **R.S. Bisht**.
- It has a **radial or oval** plan (lacking chessboard/gridiron pattern and systematic drainage), a terracotta **plough** (toy plough), mother-goddess clay figures, and good quality barley.
- It sits in the **Ghaggar** valley belt.
- **Rakhigarhi** in **Hisar, Haryana**, is the **largest** Harappan site in India.
- It has **seven mounds**. It is **not** in Uttar Pradesh.
- **Bhirrana** in Haryana is often called among the **oldest** Harappan settlements.
- **Balu** in Haryana was the 2020 match for **Haryana**.
- **Kunal** in Haryana is an Early Harappan site.
- **Farmana** in Haryana is a Mature Harappan cemetery and settlement.
- **Girawad** in Haryana is Early Harappan.
- **Mitathal** in Haryana is Harappan.
- **Ropar (Rupnagar)** on the Sutlej in Punjab was excavated by **Y.D. Sharma**.
- **Sanghol** in Punjab has Harappan and later levels.
- **Manda** on the Chenab in Jammu is the **northernmost** Harappan site (**J.P. Joshi** and Madhu Bala, 1982). It is often read as a timber-procurement outpost.
- **Bhagwanpura** in Kurukshetra, Haryana, on the Saraswati / Ghaggar belt, was excavated by **J.P. Joshi**.
- **Alamgirpur** on the Hindon in Meerut is the **easternmost** Harappan site. Gold-coated bead-like terracotta and cloth evidence are reported; pottery shows triangle, peacock, and squirrel motifs.
- **Hulas** in **Saharanpur**, Uttar Pradesh, is an eastern Harappan habitation with handmade and wheel-made pottery, chert blades, and bone points.
- **Mandi** in **Muzaffarnagar**, Uttar Pradesh, is known for a Late Harappan jewellery / treasure find (ASI). Do not confuse it with **Manda** (Jammu).
- **Sanauli** (Sonauli) in Baraut tehsil, **Baghpat**, Uttar Pradesh, yielded a large cemetery (about **125** graves, north–south). ASI 2018 work also highlights coffins, copper weapons, and **chariot** burials. It

is Late Harappan / OCP, not a Mature grid-city.

- **Dadheri** (Kotla Dadheri) in Fatehgarh Sahib, **Punjab**, is a late Harappan / **PGW** site often linked in booklet notes with later Aryan-associated pottery — not a Mature Harappan megacity.
- **Daimabad** on the Pravara in Maharashtra is the **southernmost** site.
- It yielded a bronze **chariot with charioteer**, ox, elephant, and rhinoceros.
- It also yielded bronze buffalo, elephant, and rhinoceros figures.
- **Shortughai** in Afghanistan is a **lapis lazuli** colony of the Harappans.

> **Logic:** Mandi is not Manda**. Rakhigarhi is not in Uttar Pradesh. Padri is Gujarat. Balu is Haryana. Hulas is Uttar Pradesh.
>
> [!TIP] **Exam Anchor [UPPCS Prelims 2020, Q12]:** Balu (Haryana), Manda (J&K), Padri (Gujarat), Hulas (Saharanpur, UP). *(See Q6 in Complete PYQ Bank below)*

2.4 Archaeological Sites and Excavations

Who dug what | 1826 notice to 2018 Sanauli

- **Charles Masson** in **1826** described the ruins at Harappa.
- **Alexander Cunningham** visited in the **1850s** and published a Harappa seal (**1875**), but he **did not** recognise a new civilization.
- **Daya Ram Sahni** excavated **Harappa** in **1921**.
- **R.D. Banerji** excavated **Mohenjo-daro** in **1922**.
- **John Marshall**, as ASI Director General, announced a new Bronze Age civilization in **1924** in the *Illustrated London News*.
- **M.S. Vats** worked on Harappa (**1926–27 to 1933–34**). He is also the **2020 Bhimbetka distractor**.
- **K.N. Dikshit**, Ernest Mackay, and others continued Indus work. **V.A. Smith** was **not** an Indus excavator.
- **N.G. Majumdar** discovered Chanhudaro in **1931**. He was killed in Sindh in 1938.
- **E.J.H. Mackay** directed the large Chanhudaro dig (**1935–36**).
- **Mortimer Wheeler** dug at Harappa in **1946** and introduced grid-stratigraphy.
- Wheeler **did not discover** the Indus civilization.
- He pushed an Aryan-massacre reading of Mohenjo-daro that is now **dropped**.
- **S.R. Rao** excavated the Lothal dockyard in **1955–62**.
- **B.B. Lal** excavated Kalibangan, and later Hastinapur, OCP, and Mahabharata-related archaeology.
- **R.S. Bisht** excavated Dholavira and Banawali — **not** B.K. Thapar as Dholavira excavator.
- **Y.D. Sharma** excavated Ropar (after B.B. Lal's early notice around **1950**).
- **J.P. Joshi** excavated Surkotada (and worked at Manda / Bhagwanpura).
- **Amarendra Nath** and **Vasant Shinde** worked at Rakhigarhi.
- Dates rest on stratigraphy plus **C-14**. The Mature bracket is about **2600–1900 BCE**.

- **Sanauli** in Baghpat, Uttar Pradesh, was announced by ASI in **2018**.
- It yielded wooden coffins, copper antennae swords, and **chariot burials**.
- Sanauli is Late Harappan / OCP. It is **not** a Mature grid-city with a Great Bath.
- **Sohgaura** (सोहगौरा) (Gorakhpur, UP) is a **Mauryan** copper-plate site on the Rapti — **not** a Harappan site (IAS 2019 trap with Chanhudaro / Kot Diji / Desalpur).

> **Logic:** Sahni is Harappa. Banerji is Mohenjo-daro. Wheeler is later method, not the discoverer. Cunningham saw Harappa early but missed its meaning.

2.5 Archaeological Sites of Uttar Pradesh

Eastern fringe | smaller pottery-heavy sites | where UPPCS actually scores

- Uttar Pradesh is the **eastern fringe** of Harappan culture, not the Indus core.
- **Alamgirpur** on the Hindon in Meerut marks the **eastern boundary** of the civilisation.
- **Hulas** in **Saharanpur** is an eastern Harappan habitation site in Uttar Pradesh.
- **Mandi** in **Muzaffarnagar** is in **Uttar Pradesh** (उत्तर प्रदेश). **Manda** is in Jammu.
- **Sanauli** in Baghpat (2018) has coffins, copper weapons, and **chariots**.
- Sanauli is Late Harappan / OCP. It is not a Mature urban Indus city.
- **Rakhigarhi** is in **Hisar, Haryana**, not Uttar Pradesh.
- **Kalibangan** is in Rajasthan.
- **Lothal** is in Gujarat.
- **Dholavira** is in Gujarat.
- **Surkotada** is in Gujarat.
- **Banawali** is in Haryana.
- **Balu** is in Haryana.
- **Bhirrana** is in Haryana.
- **Manda** is in Jammu and Kashmir.

> **Logic:** UP site pair = **Mandi + Hulas**; Rakhigarhi is in Haryana. Eastern boundary = **Alamgirpur**; Kalibangan and Lothal are not the eastern edge. Hulas = **Saharanpur**; Sanauli = **Baghpat**.
>
> [!TIP] **Exam Anchor [UPPCS Prelims 2025, Q87]:** Harappan sites in UP include **Mandi** and **Hulas**; Rakhigarhi is in Haryana. *(See Q1 in Complete PYQ Bank below)*
>
> [!TIP] **Exam Anchor [UPPCS Prelims 2023, Q28]:** Easternmost boundary of Harappan culture is **Alamgirpur** on Hindon River (Meerut, UP). *(See Q2 in Complete PYQ Bank below)*
>
> [!TIP] **Exam Anchor [UPPCS Prelims 2021, Q100]:** Harappan site **Mandi** is situated in Muzaffarnagar district, Uttar Pradesh. *(See Q-GC75 in Ghatna Chakra Bank below)*

> [!TIP] **Exam Anchor [UPPCS Prelims 2018, Q88]:** Harappan centres in UP = **Alamgirpur** and **Hulas** (Kalibangan = RJ, Lothal = GJ). *(See Q7 in Complete PYQ Bank below)*

2.6 Major Ports of Ancient India

Harappan coast first | later historic ports are a separate syllabus fact

- **Lothal** is the only widely accepted **artificial dockyard**, a basin with a spillway.
- A warehouse stood on a platform beside the basin.
- The site reached the Gulf of Khambhat via the Bhogavo and Sabarmati.
- **Sutkagen-dor** is a Makran coastal gate on the Arabian Sea.
- **Sotka Koh** is the neighbouring Makran coastal site.
- **Balakot** on the Pakistan coast has shell bangles and fish remains.
- **Alladino** is another Pakistan-coast craft site.
- **Kuntasi** is a Gujarat coastal workshop.
- **Bhagatrav** is a Gujarat coastal site.
- **Nageshwar** is the Gujarat **shell** industry site.
- Boat models or figures come from **Mohenjo-daro and Lothal** — not Dholavira alone.
- **Dholavira** is **not** that pair. It is inland Kutch water-harvesting.

Later ports (same syllabus bullet; not Mature Harappan cities)

- **Bharuch (Barygaza)** in Gujarat appears in the Periplus as a Roman-trade port.
- **Sopara** (शूर्पारक) is a Konkan port.
- **Muziris** in Kerala was the Roman pepper port.
- **Kaveripattinam (Puhar)** is the Sangam Chola coast.
- **Arikamedu** near Puducherry yielded Roman amphorae.
- **Tamralipti** (ताम्रलिप्ति) in Bengal served the east coast and Southeast Asia.

> **Logic:** The dockyard is **Lothal**. Boat models are **Mohenjo-daro and Lothal**. Dholavira is not the port in that question.
>
> [!TIP] **Exam Anchor [UPPCS Prelims 2022, Q68]:** Models/representations of boats excavated at **Mohenjo-daro** and **Lothal**. *(See Q3 in Complete PYQ Bank below)*

2.7 Harappan Economy

- Mixed: **agriculture + cattle + craft + long-distance trade**, organised from **cities**.
- Surplus grain in **granaries** (Harappa, Mohenjo-daro) fed specialists — bead-makers, seal-cutters, bronze-smiths, brick-makers.

- Exchange by **barter and weights**. **No coins.** Seals marked bundles, not a currency.
- Craft towns: Chanhudaro (beads), Nageshwar (shell), Lothal (beads + port).
- Do **not** import Mauryan **shreni** (श्रेणी) (guild) into IVC — that is a later institution.

> **Logic:** 2020 match: IVC = **Urban**. "Harappan coin economy" is false.

2.8 Harappan Trade

- Mesopotamian texts call the region **Meluhha**. **Dilmun** = Bahrain entrepot. **Magan** = Oman (copper).
- **Exports:** carnelian beads, cotton cloth, ivory, timber, shell, peacock (in texts).
- **Imports / raw: lapis** (Badakhshan via **Shortughai**), copper (**Khetri** / Baluchistan / Oman), tin (Iran / Afghanistan), **silver** (रूपक) (Zawar and Ajmer in Rajasthan; also Afghanistan and Iran), lead (Afghanistan / Iran / Rajasthan), gold (South India / Karnataka hinterland is often cited), shilajit (Himalaya), garnet (Gujarat).
- Steatite from Kirthar/Baluchistan; shell from Nageshwar and the Makran.
- Seals of Harappan type turn up in the Gulf and Sumer — proof of contact, not of political empire.
- Inland: ox-carts (terracotta models), river boats on Indus–Ghaggar.
- **Silver** use is early; Harappa and Mohenjo-daro are the standard silver-evidence sites.

> **Logic:** Meluhha ≈ Harappan land. Shreni is **not** a Harappan guild. First metal of mankind in booklet keys is often **copper**.

2.9 Town Planning

Grid streets | citadel west | lower town east | burnt brick

- Streets run in a **grid**, crossing at right angles.
- Main roads run north–south and east–west.
- Street width is about **3 m to 10 m**.
- The **citadel** sits on a mud-brick platform.
- It is usually **west** and higher.
- The **lower town** is **east**, larger, and used for houses and craft.
- Houses use burnt brick and rooms around a **courtyard**.
- Many houses have a private **well** and **bathroom**.
- Houses are often **two-storeyed**, with a wooden stair.
- Ground-floor **windows rarely open on the street**.
- Street corners at Mohenjo-daro show **lamp-posts**.
- **Dholavira** uses more **stone** and has **three** units: citadel, middle town, and lower town.
- Dholavira is not a photocopy of Mohenjo-daro.

- **Banawali** has **radial or oval** streets.
- Fort walls exist at Dholavira, Surkotada, and the Kalibangan citadel.
- “Unwalled peaceful villages” is too simple.
- **No** palace of a named king has been identified.

> **Logic:** The citadel is **west**. The Great Bath is **not** at every city.

2.10 Drainage System

Covered brick street drains | house bathrooms | Great Bath only at Mohenjo-daro

- Each house bathroom connected to a **covered brick street drain**.
- Soak-pits or cesspits stood at intervals.
- Manholes allowed cleaning.
- Bath floors were often made watertight with gypsum or bitumen, as at the Great Bath.
- The **Great Bath** at Mohenjo-daro is about **12 × 7 × 2.4 m**.
- It uses burnt brick plus bitumen, with steps on the north and south.
- Side rooms were used for changing.
- It is a **ritual tank**, not a sports pool.
- Kalibangan’s drainage is **not** in the same class. That is a favourite “which is NOT true of all sites” trap.

> **Logic:** The Great Bath is at **Mohenjo-daro only**.

2.11 Agriculture

Wheat and barley first | cotton early | no iron plough

- Staples are **wheat and barley**.
- Peas, sesame, mustard, dates, ragi, and melon also occur.
- **Cotton** is among the world’s **earliest**, known from cloth impressions and threads.
- **Rice** is reported at **Lothal** in Gujarat.
- Rice is also reported at **Rangpur** in Gujarat.
- Lothal / Rangpur rice is a **Harappan Gujarat** claim. **Koldihwa** rice is a **Neolithic Belan (UP)** claim.
- **Kalibangan** has a fossil **ploughed field** with a furrow grid.
- It is often cited for **double cropping**.
- **Banawali** yielded a terracotta **plough**.
- There is no iron ploughshare. The real plough was wooden.
- Irrigation used wells and flood.
- **Dholavira** used **reservoirs**.

- There is no inscribed canal-empire like Mesopotamia.
- Domestic animals include the humped bull, buffalo, sheep, goat, pig, dog, and fowl.
- The bullock pulled the plough.
- The **horse** is not a standard seal motif.
- Horse bones are only a **Surkotada** claim and remain **disputed** (onager versus horse).
- Camel bones are reported at Kalibangan.
- Elephant, rhinoceros, and tiger appear on **seals** as wild or sacred animals, not as farm stock.

> **Logic:** The ploughed field is **Kalibangan**. Cotton is yes. Horse is **not** a safe “yes.”

2.12 Craft Industries

Beads, bronze, seals, shell, terracotta | civic brick-making

- Bead materials include carnelian (heated for red), agate, steatite, lapis, and faience.
- Factory debris is densest at **Chanhudaro** and Lothal.
- Long-barrel carnelian is a Harappan export.
- **Bronze** used the **lost-wax (cire perdue)** method.
- The dancing girl is bronze.
- Daimabad bronzes include a chariot and animals.
- Pottery is wheel-thrown red ware with black-on-red painting.
- Perforated jars were used for grain or ritual.
- Seals are **steatite**, carved intaglio, then fired hard.
- Shell was used for bangles and ladles at Nageshwar, Balakot, and Lothal.
- **Faience** is a blue-green glazed paste used for jewellery.
- Terracotta includes mother goddesses, toy carts, and animals.
- **Dice and gamesmen** occur at Lothal.
- Terracotta cakes occur at Kalibangan.
- Spindle whorls show textile work. Cotton was the cloth.
- Brick-making was a civic industry with a standard size.

> **Logic:** The dancing girl is **bronze**, from Mohenjo-daro, about **10.5 cm**. Chanhudaro is beads with **no citadel**.

2.13 Religion

Reconstructed from objects | no readable scripture | no standing temple

- The **Pashupati / proto-Shiva seal** from Mohenjo-daro shows a horned yogic figure with animals around: elephant, tiger, rhinoceros, and buffalo.

- That identification is a **debate**. It is not proof of later Puranic Shaivism.
- **Mother-goddess** terracottas wear a fan-shaped headdress and jewellery and point to fertility.
- The **peepal** appears on seals.
- **Linga–yoni** stones are reported.
- The **swastika** appears on some seals.
- The **unicorn** is the most common seal animal and is mythical.
- The **humped bull** is the main real sacred animal.
- **Fire altars** occur at Kalibangan.
- Fire altars also occur at Lothal.
- Fire altars also occur at Banawali.
- Kalibangan fire altars are a **Harappan** feature. Vedic yajna is a **later textual** ritual system.
- The Great Bath points to ritual purity and a water cult.
- Burials are usually extended inhumation, often north–south, with pottery and ornaments.
- **Fractional** burial and **urn** burial occur in late levels.
- **Double burial** occurs at Lothal.
- Harappa has **coffin** burials in cemetery **R-37**.
- There are **no** royal pyramids.
- Idol / figurine worship is often treated as beginning in the **pre-Aryan / Harappan** horizon (mother goddess and related terracottas).
- Harappan religion is often called **secular** in the sense that no theocratic temple-state is proved. Religious element is present but does not dominate like later temple cultures.
- There are **no** large free-standing temples of later Hindu type.
- **Cemetery H** at Harappa is Late Harappan, with painted urns and fractional burial.
- Cemetery H is a **Late Harappan** burial phase at Harappa. It does **not** prove an Aryan invasion by itself.

> **Logic:** “A Hindu temple was excavated at Mohenjo-daro” is false. Pashupati is an **interpretation**. No large temples; mother goddess and proto-Shiva seal are the usual keys.

2.14 Script

Undeciphered Harappan signs | short texts | no Rosetta Stone

- The Harappan script is **undeciphered**.
- There are about **400–600** signs.
- Most texts are only **5–6** signs long.
- The longest seal-text is often cited at about **26** signs (Harappa).
- The **Dholavira signboard** carries about **10** large signs.
- Writing appears on seals, copper tablets, pottery, and bangles.
- There is no long literary book.

- Direction is generally **right to left**. Some lines are **boustrophedon** (alternate like a ploughed field).
- There is no bilingual key like a Rosetta Stone.
- Dravidian and Indo-Aryan readings both exist. **Neither is proved**.
- The script appears fully in the **Mature** phase and thins out in Late Harappan.

> **Logic:** "Script has been read as Sanskrit" = false. Signboard = **Dholavira**.

2.15 Seals

Steatite stamps | trade and identity | unicorn is the common animal

- Seals are square or rectangular **steatite**, about **2–4 cm**.
- They are carved **intaglio**, then fired.
- An animal sits below. **Script** sits above.
- The reverse has a boss for a cord.
- Clay **sealings** closed packages and doors.
- The **unicorn** is the **most common** animal type.
- Next comes the humped bull. Buffalo, elephant, rhino, tiger, goat, sheep, and rare composite monsters also appear.
- The **cow** is **not** depicted on seals or terracotta art.
- The **horse** is **not** a standard seal animal.
- The **peepal** tree appears on seals.
- The **Pashupati** seal is the religious celebrity. It is **not** the statistical majority.
- Function is identity, cargo control, and amulet use. Seals are **not coins**.
- Harappan seals in Mesopotamia prove **trade**, not conquest.
- Knowledge of the culture rests mainly on **archaeological excavation**. Seals are the richest single artefact class, but the unread script is not yet a readable history.

> **Logic:** Material = **steatite**. Unicorn ≠ horse. Cow ≠ seal animal.

2.16 Weights and Measures

Cubical stone weights | binary then decimal series | not coined money

- Weights are cubical stones of **chert / jasper / black stone**.
- The same system runs from Afghanistan to Maharashtra.
- The series runs **1, 2, 4, 8, 16, 32, 64…**, then bigger decimal jumps (**160, 200, 320, 640…**).
- The key ratio to remember is **16**.
- The smallest unit is often about **0.85 g**.
- Length is measured with shell or ivory **scales** at Mohenjo-daro and Lothal.

- Cubit-like units are debated.
- There are **no** gold punch-marked coins here. Those belong to later Janapada / Mahajanapada times.

> **Logic:** Weights = **stone cubes**, not minted coins.

2.17 Dholavira

Khadir island, Rann of Kutch, Gujarat | R.S. Bisht | UNESCO WHS 2021

- Dholavira is divided into three parts — **citadel, middle town, and lower town** — a layout unique among the big Harappan cities.
- Builders used more **stone** here than in the all-brick Punjab–Sindh cities.
- Elaborate **water harvesting** includes check dams, storm channels, and about **16 reservoirs**.
- Rock-cut and rock-lined reservoirs are a Dholavira specialty (booklet depths about **7 m** and lengths about **79 m** in older notes).
- A famous **signboard** of about **10** large Harappan signs stood near the northern gate — the biggest public inscription yet found.
- Street drainage here is **not** a clone of Mohenjo-daro's covered-brick textbook system.
- Trade ran inland across the Rann of Kutch; this site is **not** Lothal's dockyard and is **not** the 2022 boat-model answer pair.

> **Logic:** UNESCO **2021** = Dholavira. Water ≠ dock.

2.18 Rakhigarhi

Hisar, Haryana (Ghaggar–Hakra) | largest Harappan site in India | 7 mounds | ~550 hectares often cited

- Sequence from Early to Mature; planned housing, crafts, burials.
- **Amarendra Nath** (ASI, 1997–2000) excavated Rakhigarhi in a major campaign.
- **Vasant Shinde** (Deccan College) led later work and DNA-related publications from the site.
- ASI is still digging; 2026 reports sent skeletons for DNA and facial reconstruction under a multi-year programme.
- DNA papers (Shinde et al.) are used in "Aryan" debates — for Prelims: site = **Haryana**, **not** the eastern end, **not UP**.
- Budget **2026**: listed with Lothal among 15 experiential archaeological destinations.

> **Logic:** Rakhigarhi is in **Haryana**, not Uttar Pradesh. Eastern boundary remains **Alamgirpur**.

2.19 Lothal Dockyard

Ahmedabad district belt, Gujarat | S.R. Rao 1955–62

- The artificial **dockyard** / basin measures about **218 × 37 m** and has a spillway; this is the principal Harappan **port** card.
- A warehouse stood on a mud-brick platform beside bead and copper workshops, a **fire altar**, rice husk, dice/gamesmen, and a **double burial**.
- **Boat models** from Lothal are paired with Mohenjo-daro in the 2022 stem.
- An **ivory scale** is a Lothal booklet fact (shell and ivory scales are also taught for Mohenjo-daro).
- The town has an acropolis and lower town and linked to the Gulf of Khambhat by the Bhogavo.
- Lothal is **not** in Uttar Pradesh and is **not** Dholavira.

> **Logic:** If the option says dockyard / port / boats / ivory scale, think **Lothal** first.

2.20 Kalibangan

Hanumangarh, Rajasthan | Ghaggar | B.B. Lal

- A pre-Harappan **ploughed field** with criss-cross furrows is the unique agriculture fact of Kalibangan.
- **Fire altars** stand on the citadel and in the lower town, with terracotta cakes in the ritual set.
- The plan has a citadel and lower town; early levels use mud-brick, and **drainage is poorer** than at Mohenjo-daro.
- Cylindrical seals differ from the common square steatite type, and **camel** bones are reported.
- Kalibangan is **not in Uttar Pradesh** (a frequent 2018 distractor).

> **Logic:** Ploughed field + fire altars = **Kalibangan**. Great Bath ≠ here.

2.21 Banawali

Fatehabad, Haryana | Ghaggar–Hakra | R.S. Bisht

- Banawali shows Early and Mature Harappan levels and is **fortified**.
- Streets run in a more **radial / oval** pattern than a strict grid.
- Finds include a terracotta **plough**, barley, beads, and fire-altar reports.
- The site is in **Haryana**, not Rajasthan (do not swap it with Kalibangan) and is not a dockyard.

2.22 Surkotada

Kutch, Gujarat | J.P. Joshi

- Small **fortified** citadel plus residential annexe.
- **Horse bones** at Surkotada are still **disputed** (true horse vs onager/wild ass). Papers treat the horse as **unsettled**.
- Not a dockyard; not in Rajasthan.

2.23 Daimabad

Ahmednagar / Pravara, Maharashtra | southernmost Harappan-related site

- Late Harappan / Chalcolithic overlap (Jorwe).
- Bronze hoard: **chariot**, water buffalo, elephant, rhinoceros.
- These bronzes are **not** the same as Sanauli's wooden/copper **chariot burials** in Baghpat.
- **Maharashtra**, not Gujarat.

2.24 Copper Hoard Culture

Stray copper tools and figures in the Gangetic doab | usually paired with OCP | not Harappan workshop bronze

- The **Copper Hoard Culture** is a scatter of **copper** objects across the **Ganga–Yamuna doab** and neighbouring belts: **anthropomorphs**, **harpoons**, **antennae swords**, **celts**, and axes.
- The largest hoard is **Gungeria** in **Balaghat**, Madhya Pradesh.
- These objects are usually discussed with **Ochre Coloured Pottery (OCP)** sites. They are **not** the same as the urban bronze crafts of Mature Harappan **Mohenjo-daro** or **Chanhudaro**.
- **Sanauli** (Baghpat, UP) copper weapons and chariot burials belong to the Late Harappan / OCP conversation in the upper doab. They are **not** Gujarat dockyard bronzes.
- The culture is rural and pre-urban. It sits **before** PGW and long before NBPW cities.
- Trap: "anthropomorph" on a paper usually means this **Gangetic copper hoard** type, not a Harappan seal figure.

> **Logic:** Anthropomorph = Gangetic Copper Hoard shape. Not a Gujarat dockyard craft.

2.25 Ochre Coloured Pottery (OCP)

Late rural doab culture | ochre wash on pottery | bridge between Late Harappan east and PGW

- **Ochre Coloured Pottery** gets its name from a red ochre wash that rubs off on the fingers.
- The main belt is **western UP, Haryana, and Rajasthan**: **Hastinapur** (हस्तिनापुर), **Bahadarabad**, **Lal Qila**, **Bargaon**, and **Atranjikhera**.
- **B.B. Lal** excavated OCP at **Hastinapur**. That dig is famous in the Mahabharata archaeology debate, but OCP alone does **not** prove the epic war.

- The date-band is roughly the **late 3rd to 2nd millennium BCE**. It overlaps the **eastern fringe** of Late Harappan decline and comes **before PGW** in stratigraphy.
- Settlements are **rural**, not planned Harappan cities. **Copper Hoards** often appear nearby.
- OCP is **not** Lothal red ware. It is **not** megalithic black-and-red ware of the south.
- Sequence memory for the doab: **Late Harappan / OCP → PGW → NBPW**.

> **Logic:** OCP = **doab** (दोआब), not Lothal red ware. Hastinapur OCP ≠ automatic "Mahabharata proved."

UP Focus (once)

- **Alamgirpur** (Meerut) — eastern boundary of the IVC in India.
- **Hulas** (Saharanpur) — 2018, 2025.
- **Mandi** (Muzaffarnagar) — 2021, 2025.
- **Sanauli** (Baghpat) — 2018 chariots, Late Harappan/OCP.
- OCP belt = western UP doab (Hastinapur).
- Rakhigarhi is **not** this list.